

Cloud-based EDA tools available

1. Nimbic nCloud. It's Nimbic's cloud solution. It consists of a Web portal to access the services, and infrastructure to host and deliver the software that constitutes nWave and nApex.

2. Synopsys VCS. It is a functional verification solution that provides high-performance simulation engines, constraint solver engines, Native Testbench support, broad SystemVerilog support, verification planning, coverage analysis and closure, and an integrated debug environment.

3. Cadence Hosted Design Solutions. It applies the same concepts as in the traditional SaaS application stack, and adapts them to the demanding data and processing requirements of leading-edge semiconductor design.

4. SiCAD's Virtual Design Center. It has been specifically architected to easily scale compute resources and EDA tools for ASIC/SoC design. It allows customers to configure and deploy compute and design resources when needed, for the duration they want.

ing data stored on the local server as opposed to leaving it on the cloud.

Dr Raul Camposano, CEO, Nimbic, explains in his whitepaper *Electronic Design in the Cloud*, "A common connection of 10 Mbps allows up to 75 MB of data transfer in one minute (theoretical maximum). The user has to decide if it is practical to transfer design data back and forth to the cloud for a given application. In the case of nWave for example, the design data and results are only a few megabytes, allowing for a transfer of data to/from the cloud in seconds. Compared to the compute time required to solve a problem, often in the hours or days, this is negligible. For other applications that require more data, an alternative is to leave the data in the cloud, using a service such as Amazon's S3."

However, not everyone is open to the idea of leaving data on the cloud, especially when considering data security.

Data security. Data security is a sensitive topic. With the proliferation of cloud-based solutions in a lot of cloud-based applications, there have been numerous cases of personal data being misused for other purposes.

Could a variation of the same problems crop up while designing on the Internet?

"Major issue I foresee is design sanctity and security. When the design data has to be on the cloud, as a design house, I would want my data to be absolutely secure and untouchable as it is a core part of the business. And since cloud model for other services has still not proven its capability in all aspects, its viability for EDA is under question," says Anand.

Is it really going to be cheaper?

Overall, there will be almost zero capital expenses (CAPEX) as all of the cost would be under operational expenses (OPEX).

"The biggest drawback is that these technical and cost efficiencies are achieved partly through standardisation. If you need a highly customised flow, a SaaS-style model might not be the best fit for your project," explains Drenan.

While EDA in the cloud may reduce licensing cost (as it changes from one-time to subscription-/usage-

Will the SaaS model work for everyone?

"Working model for cloud has been SaaS, which has been useful in many transactional operation cases. However, EDA is not transaction-based use; and therefore either some radical operational change would be necessary to make this successful or the model would remain just another exotic offering by EDA tool providers"

—Anand Tamboli, managing director, Knewron

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—Larry Drenan, group director for services, Cadence Design Systems, Inc.

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